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NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fourth Semester B.E. (AIML/CCE) (Credit System) Degree Examinations****Supplementary Examinations - December 2025****21AM403 / 20AM404 / 21CC403 / 20CC404 - DATABASE SYSTEMS**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

	Marks	BT*	CO*	PO*
1. a) Elaborate on the characteristics of the database approach.	8	L*1	1	1
b) With a neat diagram, explain the three-schema architecture.	6	L1	1	1
c) With suitable illustrations, explain the different types of attributes.	6	L2	1	1
2. a) Write an E-R diagram for the Hospital Management System. Write necessary assumptions made.	10	L2	1	1
b) Elaborate on the characteristics of the relations.	10	L2	2	1
3. a) Write the relational algebra expressions for the following on the database schema given. CATALOG(Cno, Ctitle) STUDENTS(Sid, Fname, Lname, Minit) COURSES(Term, Sec_no, Cno, A, B, C, D) ENROLLS(Sid, Term, Sec_no)				

Queries:

- Retrieve the names of the students enrolled in the Automata class during the term 2009.
 - Retrieve the sid values of students who have enrolled in 21CS403 and 21CS404.
 - Retrieve the names of the students who have not enrolled in any class.
 - Retrieve the names of the students who have enrolled in all the courses in the CATALOG table.
- | | | | |
|----|----|---|---|
| 10 | L3 | 2 | 2 |
|----|----|---|---|
- Differentiate between inner join and outer join with suitable examples.
 - What are aggregate functions? How we use aggregate functions in relational algebra expressions?
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|---|----|---|---|
| 6 | L1 | 2 | 2 |
| 4 | L1 | 2 | 2 |

Unit – II

- Write the E-R to relational mapping algorithm and explain each step with suitable examples.
 - How to specify constraints in SQL? Give suitable examples.
 - Write SQL statements for the following queries using the schema given.
EMPLOYEE(SSN, Name, Address, DOB, Salary, SuperSSN, DNo)
DEPARTMENT(DNumber, DName, MGRSSN)
PROJECT(PNumber, PName, Plocation, DNum)
WORKS_ON(SSN, PNo, Hours)
 - Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees.
 - Retrieve the names of all managers.
 - Retrieve the total number of employees in each department.
 - Retrieve the employees who are working on ProductX project.
- | | | | |
|----|----|---|---|
| 12 | L2 | 3 | 2 |
| 8 | L1 | 3 | 2 |
| 10 | L3 | 3 | 2 |

P.T.O.

- b) What are triggers? Write a trigger for the deletion operation on the EMPLOYEE table. 5 L2 3 2
- c) Briefly elaborate the use of views. 5 L2 3 2
6. a) What are stored procedures? How it is helpful? Elaborate with suitable examples. 10 L2 4 2
- b) Which normal form is based on full functional dependence? Elaborate on the normal form which is based on this with suitable examples. 10 L2 4 2

Unit – III

7. a) Discuss the minimal cover algorithm with examples. 10 L2 5 2
- b) Highlight the ACID properties of a transaction. 5 L1 5 2
- c) Briefly explain the fourth normal form with suitable examples. 5 L2 5 2
8. a) Explain with the sample code, the two-phase locking protocol. 10 L2 5 2
- b) What is serializable of a schedule? Write an algorithm to find whether the given schedule is serializable or not. Also, check whether the following schedule is serializable or not.

	Transaction T_1	Transaction T_2	Transaction T_3
Time ↓	read_item(X); write_item(X); read_item(Y); write_item(Y);	read_item(Z); read_item(Y); write_item(Y); read_item(X); write_item(X);	read_item(Y); read_item(Z); write_item(Y); write_item(Z);

10 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
